Chapter-1

Two-Dimensional Motion and Newton's Law

Textbook: Fundamentals of Physics by Halliday, Resnick & Walker [10th edition]

Topics to be covered:

4-1: Position and Displacement. 4-2: Average Velocity and Instantaneous Velocity. 4-3: Average Acceleration and Instantaneous Acceleration. 4-4: Projectile motion. 5-1: Newton's First and Second Laws, 5-2: Some particular forces (Gravitational force, Weight, Normal force, Tension), 5-3: Applying Newton's law.

Theory

1. Show that the trajectory of a projectile is a parabola.
2. Consider a projectile with projection angle θ_o and initial velocity v_o. The time to reach the maximum height H is t and the total time of flight to cover the range R is T and the final velocity is v. Derive the following expressions for that projectile.

   (a) $t = \frac{v_o\, Sin\theta o}{g}$ (b) $T = \frac{2v_o Sin\theta o}{g}$ (c) $H = \frac{{v_o}^2 Sin^2\theta o}{2g}$ (d) $R = \frac{{v_o}^2 Sin2\theta o}{g}$
3. Show that for the maximum range of a projectile, the projection angle is 45°.
4. State and explain Newton's Laws of Motion.
5. Define: (a) Force, (b) Gravitational force, (c) Weight, (d) Normal force and (e) Tension.

Motion in 2D and 3D: Projectile Motion (Chapter-4)

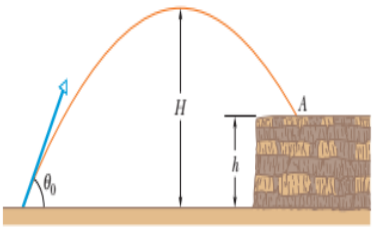
1. [23] A projectile is fired horizontally from a gun that is 45.0 m above flat ground, emerging from the gun with a speed of 250 m/s. (a) How long does the projectile remain in the air? (b) At what horizontal distance from the firing point does it strike the ground? (c) What is the magnitude of the vertical component of its velocity as it strikes the ground?

2. [Sample problem-4.04(b)] A rescue plane flies at 198 km/h and constant height h = 500 m toward a point directly over a victim, where a rescue capsule is to land. As the capsule reaches the water, (a) what is its velocity in unit-vector notation, (b) what are the magnitude and direction of velocity?

3. [21] A dart is thrown horizontally with an initial speed of 10 m/s toward point *P*, the bull's eye on a dart board. It hits at point *Q* on the rim, vertically below *P*, 0.19 s later. (a)What is the distance *PQ*? (b) How far away from the dart board is the dart released?

4. [32] You throw a ball toward a wall with a speed of 25.0 m/s and at an angle of40.0º above the horizontal. The wall is 22.0m from the release point of the ball. (a) How far above the release point does the ball hit the wall? (b) What are the horizontal and vertical components of its velocity as it hits the wall?

5. [28] A stone is projected at a cliff of height *h* with an initial speed of 42.0 m/s directed at angle θ_0=60.0° above the horizontal. The stone strikes at *A*, 5.50 s after launching. Find (a) the height *h* of the cliff, (b) the speed of the stone just before impact at *A*, and (c) the maximum height *H* reached above the ground.



6. [26] A stone is catapulted at time t= 0, with an initial velocity of magnitude 20.0 m/s and at an angle of 40.0° above the horizontal. What are the magnitudes of the (a) horizontal and (b) vertical components of its displacement from the catapult site at t =1.10 s and at t= 1.80 s?

7. [25] The current world-record motorcycle jump is 77.0 m, set by Jason Renie. Assume that he left the take-off ramp at 12.0º to the horizontal and that the take-off and landing heights are the same. Neglecting air drag, determine his take-off speed.

Application of Newton's Laws of Motion (Chapter-5)

8. [1] Only two horizontal forces act on a 3.0 kg body that can move over a frictionless floor. One force is 9.0 N, acting due east, and the other is 8.0 N, acting 62° north of west. What is the magnitude of the body's acceleration?

9. [Sample Problem-5.07] A constant horizontal force *F* of magnitude 20 N is applied to block *A* of mass m_A= 4.0 kg, which pushes against block *B* of mass m_B = 6.0 kg. The blocks slide over a frictionless surface, along an *x* axis. (a) What is the acceleration of the blocks? (b) What is the (horizontal) force on block *B* from block *A?*

10. [25] A spacecraft with a large sail is pushed by sun light. Suppose that the spacecraft has a mass of 900 kg and receives a push of 20 N. (a) What is the magnitude of the resulting acceleration? If the craft starts from rest, (b) how far will it travel in 1 day and (c) how fast will it then be moving?

11. [27] An electron with a speed of 1.2×10^7 m/s moves horizontally into a region where a constant vertical force of 4.5×10^{-16} N acts on it. The mass of the electron is 9.11×10^{-31} kg. Determine the vertical distance the electron is deflected during the time it has moved 30 mm horizontally.

12. [33] An elevator cab and its load have a combined mass of 1600 kg. Find the tension in the supporting cable when the cab, originally moving downward at 12 m/s, is brought to rest with constant acceleration in a distance of 42 m.